

|           | $\sqrt{0.12 \times 0.08^2 \times (2\pi)^2}$                                                                                                                                                                                                                                                                                     |                            |                                |                            |                                |       |      |      |     |       |      |   |     |       |   |      |      |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|--------------------------------|----------------------------|--------------------------------|-------|------|------|-----|-------|------|---|-----|-------|---|------|------|
| 3(a)      | Change in the internal energy of the system is equal to the summation of the net heat supplied to the system and net work done on the system.                                                                                                                                                                                   |                            |                                |                            |                                |       |      |      |     |       |      |   |     |       |   |      |      |
| 3(b)(i)   | $pV = nRT$<br>$n = \frac{pV}{RT} = \frac{2.4 \times 10^5 \times 5 \times 10^{-4}}{8.31 \times 290} = 0.05 \text{ mol}$                                                                                                                                                                                                          |                            |                                |                            |                                |       |      |      |     |       |      |   |     |       |   |      |      |
| 3(b)(ii)1 | $WD = p\Delta V = 2.4 \times 10^5 \times (14.4 - 5.0) \times 10^{-4} = 225.6 = 230 \text{ J}$                                                                                                                                                                                                                                   |                            |                                |                            |                                |       |      |      |     |       |      |   |     |       |   |      |      |
| 3(b)(ii)2 | In one complete cycle XYZX, the change in internal energy is 0 J.                                                                                                                                                                                                                                                               |                            |                                |                            |                                |       |      |      |     |       |      |   |     |       |   |      |      |
| 3(b)(iii) | <table><tr><th>Change</th><th>Work done on gas /J</th><th>Heating supplied to gas /J</th><th>Increase in internal energy /J</th></tr><tr><td>X → Y</td><td>-230</td><td>+570</td><td>340</td></tr><tr><td>Y → Z</td><td>+540</td><td>0</td><td>540</td></tr><tr><td>Z → X</td><td>0</td><td>-880</td><td>-880</td></tr></table> | Change                     | Work done on gas /J            | Heating supplied to gas /J | Increase in internal energy /J | X → Y | -230 | +570 | 340 | Y → Z | +540 | 0 | 540 | Z → X | 0 | -880 | -880 |
| Change    | Work done on gas /J                                                                                                                                                                                                                                                                                                             | Heating supplied to gas /J | Increase in internal energy /J |                            |                                |       |      |      |     |       |      |   |     |       |   |      |      |
| X → Y     | -230                                                                                                                                                                                                                                                                                                                            | +570                       | 340                            |                            |                                |       |      |      |     |       |      |   |     |       |   |      |      |
| Y → Z     | +540                                                                                                                                                                                                                                                                                                                            | 0                          | 540                            |                            |                                |       |      |      |     |       |      |   |     |       |   |      |      |
| Z → X     | 0                                                                                                                                                                                                                                                                                                                               | -880                       | -880                           |                            |                                |       |      |      |     |       |      |   |     |       |   |      |      |
| 4(a)      | The principle of superposition states that when two or more travelling waves                                                                                                                                                                                                                                                    |                            |                                |                            |                                |       |      |      |     |       |      |   |     |       |   |      |      |